

Missing Averages



A. Find the mode, median, mean and range for each of the following sets of data.

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| 1). 7 kg, 3 kg, 9 kg, 8 kg, 3 kg. | 2). 6 ml, 9 ml, 5 ml, 2 ml, 13 ml. |
| 3). 12 min, 7 min, 17 min, 12 min, 7 min. | 4). 9 cm, 3 cm, 5 cm, 7 cm, 3 cm, 12 cm. |
| 5). 17 m, 13 m, 16 m, 19 m, 8 m, 11 m. | 6). 42 g, 16 g, 28 g, 15 g, 33 g, 19 g. |

B. Given the information for each set of data, find the missing number. All numbers are positive whole numbers, For some there may be **more than one** answer. Give all possible answers.

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| 1). x , 3, 16, 5. The mode = 5. | 2). 7, 9, m , 5. The mode = 7. |
| 3). 9, g , 12, 7, 15. The mode = 9. | 4). 13, 15, 15, 13, a . The mode = 13. |
| 5). 7, n , 14, 12. The range = 10. | 6). 23, 15, h , 24. The range = 12. |
| 7). 13, 20, 15, 18, y . The range = 8. | 8). t , 54, 41, 39, 47. The range = 18. |
| 9). 11, 19, 2, u , 19. The median = 11. | 10). 9, k , 14, 12, 17. The median = 13. |
| 11). 14, 16, 18, 10, 11, g . The median = 15. | 12). 20, 25, 21, m , 30, 45. The median = 26. |
| 13). 5, 7, e , 9. The mean = 6. | 14). 12, 8, 14, r . The mean = 11. |
| 15). p , 17, 19, 12, 14. The mean = 15. | 16). 26, 13, w , 18, 16. The mean = 18. |
- 17). The mean of three numbers is 8. Two of the numbers are 8 and 12. What is the third number?
- 18). The mode of four numbers is 5. Three of the numbers are 6, 9 and 5. What is the fourth number?
- 19). The range of four numbers is 12. Three of the numbers are 7, 15 and 13. What two numbers could be the fourth number?
- 20). The median of five numbers is 9. Four of the numbers are 9, 12, 7 and 8. What numbers could be the fifth number?

C.

- 1). a). The mean of a set of 6 numbers is 12. What is the sum of the 6 numbers?
b). The sum of a set of 9 numbers is 45. What is the mean of the set of 9 numbers?
c). The sum of a set of numbers is 108 and the mean 12. How many numbers are in the set?
- 2). The mean price for 5 items is \$8.00. Four of the items cost \$9.00, \$7.50, \$7.00 and \$6.00. How much did the fifth item cost?
- 3). The mean of three numbers is 11. One of the numbers is 14.
a). Write what the other two numbers could be.
b). Write another pair of numbers that are possible.
c). If the two numbers are a and b , form an equation to show the relationship.
- 4). The mean of four numbers is 9. Two of the numbers are 3 and 7.
a). Write what the other two numbers could be.
b). Write another pair of numbers that are possible.
c). If the two numbers are x and y , form an equation to show the relationship.
- 5). Find 5 numbers that have a range of 5, a mean of 15 and a median of 16.
- 6). a). Year 8 hockey team score these goals in matches: 15, 7, 11, 15, 11, 1, 17, 0, 10.
i). Find the mode, ii). find the median.
b). Year 9 hockey team score these goals in matches: 5, 10, 7, 10, 5, 13, 17, 9, 17.
i). Find the mode, ii). find the median.
c). Using then mode and median, explain which team you think is the best.
d). The Year 10 netball team also play 9 matches. They have the same mode as Year 8, the same median as Year 9, but scored less goals altogether than either team. What could be their monthly goals totals?
- 7). a). The mean of a set of numbers is 7 and the range 9. The numbers are all doubled. What is the new mean and the new range?
b). The mode of a set of numbers is 20 and the range 6. The numbers are halved. What is the new mode and the new range?

